

Orion-Bix: Bi-Axial Attention for Tabular In-Context Learning

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Abstract

Tabular data drive most real-world machine learning applications, yet building general-purpose models for them remains difficult. Mixed numeric and categorical fields, weak feature structure, and limited labeled data make scaling and generalization challenging. To this end, we introduce Orion-Bix, a tabular foundation model that combines biaxial attention with meta-learned in-context reasoning for few-shot tabular learning. Its encoder alternates standard, grouped, hierarchical, and relational attention, fusing their outputs through multi-CLS summarization to capture both local and global dependencies efficiently. A label-aware in-context learning (ICL) head adapts on the fly and scales to large label spaces via hierarchical decision routing. Delivered as a scikit-learn-compatible foundation model, it outperforms gradient-boosting baselines and remains competitive with state-of-the-art tabular foundation models on public benchmarks, showing that biaxial attention with episodic meta-training enables robust, few-shot-ready tabular learning.

CCS Concepts

• **Computing methodologies** → **Supervised learning by classification**; **Neural networks**; **Machine learning**.

Keywords

Tabular Foundation Models, Transformer-based Models, Bi-Axial Attention, Tabular In-Context Learning

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1 Introduction

Tabular data remain the most common form of real-world data, spanning domains such as healthcare, finance, and scientific research. Despite major advances in deep learning for language processing and vision [7], gradient-boosted trees (GBTs) remain state-of-the-art for tabular prediction. Recently, tabular in-context learning (ICL) has emerged, adapting the ICL paradigm—central to large language

models—to tabular data, enabling pretraining across diverse tables and rapid task adaptation without gradient updates.

TabPFN [2] pioneered this approach by meta-training a transformer on synthetic datasets generated via structural causal models. Its encoder-decoder design lets test samples attend to training examples for zero-shot prediction, but alternating column- and row-wise attentions make large training sets computationally costly. TabDPT [4] achieves comparable performance using similarity-based retrieval, though its diffusion process adds overhead. TabPFN-v2 [3] extended row-based encoding to datasets exceeding 10,000 samples. TabICL [5] further refined this with a table-native transformer comprising column embeddings, row interactions, and an ICL head. Its SetTransformer-based column encoder and label-aware in-context learner achieve state-of-the-art results, but its row encoder (`tf_row`) applies a single homogeneous attention over all features, ignoring local groupings, multi-scale interactions, and structured aggregation. This limitation is pronounced in high-dimensional, heterogeneous tables where features naturally form semantic groups (e.g., demographics, vitals, labs) and dependencies span multiple scales. Additionally, TabICL treats synthetic tables as individual supervised tasks rather than support/query episodes, producing an implicit few-shot signal that may misalign with test-time objectives.

To address these limitations, we introduce **Orion-Bix**¹, which enhances TabICL with two key improvements: (1) a *biaxial row encoder* replacing `tf_row` with complementary attention modes—standard cross-feature, grouped, hierarchical, and relational—aggregated via multiple CLS tokens, explicitly modeling local groups, coarse-scale interactions, and global patterns; and (2) a *meta-learning training regime* that constructs support/query episodes from synthetic tables, optionally using kNN-based support selection, aligning training with few-shot test-time objectives. Orion-Bix preserves TabICL’s strengths in column-wise SetTransformer embeddings and label-aware ICL while addressing its row-level and training limitations.

2 Orion-Bix: Proposed Approach

Orion-Bix retains two core TabICL [5] components: the SetTransformer column encoder and the label-aware in-context learner, while introducing two major changes: (1) a biaxial row encoder that replaces the single-stage `tf_row` and provides structured multi-scale feature reasoning, and (2) a meta-learning training regime that constructs explicit support/query episodes from synthetic tables, optionally using kNN-based support selection.

¹The code and model checkpoint are available at <https://github.com/Lexsi-Labs/Orion-Bix>



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2.1 Problem Setting and Data Representation

We consider a tabular task with a table $X \in \mathbb{R}^{n \times m}$ and labels $y \in \{0, \dots, C-1\}^n$. Rows are split into a *support set* of size n_{train} and a *query set* of size n_{test} . During inference and meta-training, Orion-Bix receives X and the support labels

$$y_{\text{train}} = y_{1:n_{\text{train}}}.$$

For batched tasks,

$$X \in \mathbb{R}^{B \times n \times m}, \quad d \in \mathbb{N}^B,$$

where d_b records the number of active (non-padding) features for task b . The objective is *tabular in-context learning*: predict all query labels conditioned solely on support rows and their labels.

2.2 Column-wise Embedding

We adopt the original TabICL [5] column-wise embedder to map each feature to a d -dimensional representation per cell. The embedder uses a *Set Transformer*, TF_{col} , that treats each column as a permutation-invariant set of values across all rows.

Concretely, given the input table $X \in \mathbb{R}^{B \times n \times m}$, the embedder:

- Transposes to (B, m, n) and treats each of the m columns as a set of n values.
- Applies TF_{col} to each column set, producing per-cell weights and biases: $W, B \in \mathbb{R}^{B \times m \times n \times d}$.
- Forms the column embeddings via element-wise affine transformation: $E = W \odot X + B \in \mathbb{R}^{B \times n \times m' \times d}$, where $m' = m + N_{\text{CLS}}$ includes reserved slots for CLS tokens.

This stage is responsible for distribution-aware, type-agnostic feature normalization and preserves TabICL's strengths in modeling per-column statistics.

2.3 Biaxial Row Encoder: Multi-Scale Feature Interactions

The column embeddings $E \in \mathbb{R}^{B \times n \times m' \times d}$ provide per-cell feature representations. The row encoder must transform these into row-level summaries suitable for in-context reasoning. TabICL's row encoder applies a single Transformer stack over the feature dimension, treating all columns as a flat sequence. This forces one homogeneous attention pattern to capture (i) local dependencies within semantically related feature subsets, (ii) long-range cross-column interactions, and (iii) global aggregation into a single row vector.

In high-dimensional, heterogeneous tables, this uniform treatment introduces three limitations: (i) **no explicit feature grouping**, despite tables containing natural substructures (e.g., demographics, vitals, lab tests); (ii) **weak multi-scale structure**, since local and global relationships must be modeled through the same attention mode; and (iii) a **representation bottleneck**, as collapsing a row into a single embedding mixes distinct informational modes.

To overcome this, Orion-Bix replaces the flat row encoder with a *biaxial attention module* that applies multiple, specialized feature-space attentions and aggregates their outputs through multi-CLS tokens, enabling structured, multi-scale row representations.

2.3.1 Input Reshaping. Starting from the column embeddings $E \in \mathbb{R}^{B \times n \times m' \times d}$, we reshape to treat each row's features as a sequence:

$$E \rightarrow X' \in \mathbb{R}^{(B \cdot n) \times m' \times d},$$

where each element $X'_i \in \mathbb{R}^{m' \times d}$ represents one row's sequence of feature embeddings.

2.3.2 Bi-Axial Attention Block. Each BiAxialAttentionBlock applies four attention modules in sequence over the feature axis:

- (1) **Standard attention:** full cross-feature self-attention capturing generic dependencies:

$$X_1 = \text{Attn}_{\text{std}}(X_0, X_0, X_0),$$

- (2) **Grouped attention:** local attention within G feature groups of size $\lfloor m'/G \rfloor$. Within each group, we apply a local attention:

$$X_2 = \text{Attn}_{\text{group}}(X_1),$$

- (3) **Hierarchical attention:** coarse-scale interactions between two large partitions of the feature set $X_2 = [X_2^{(1)}; X_2^{(2)}]$:

$$\hat{X}_2^{(1)} = \text{Attn}_{\text{hier}}(X_2^{(1)}, X_2^{(2)}, X_2^{(2)}),$$

$$\hat{X}_2^{(2)} = \text{Attn}_{\text{hier}}(X_2^{(2)}, \hat{X}_2^{(1)}, \hat{X}_2^{(1)}).$$

- (4) **Relational attention:** a second full self-attention over already structured features:

$$X_3 = \text{Attn}_{\text{rel}}(\hat{X}_2, \hat{X}_2, \hat{X}_2),$$

2.3.3 Multi-CLS Aggregation. Each of the previous blocks introduces N_{CLS} learnable class tokens that attend over the feature sequence:

$$\text{CLS}' = \text{CLSAttn}(\text{CLS}, X_3) \in \mathbb{R}^{(B \cdot n) \times N_{\text{CLS}} \times d}.$$

Stacking L_{row} blocks refines these summaries. After flattening,

$$R \in \mathbb{R}^{B \times n \times (N_{\text{CLS}} \cdot d)}$$

is passed to the in-context learner. Multiple CLS tokens prevent information bottlenecks and allow Orion-Bix to jointly model local interactions within feature groups and global relational patterns across all features.

2.4 In-Context Learner

The in-context learner mirrors TabICL:

- It receives row representations $R \in \mathbb{R}^{B \times n \times d_{\text{row}}}$ and support labels $y_{\text{train}} \in \{0, \dots, C-1\}^{B \times n_{\text{train}}}$.
- Support labels are one-hot encoded, linearly projected, and added to support row embeddings.
- A Transformer with a split mask enforces: support rows attend to support only; query rows attend to support but not to each other.
- An MLP produces logits in $\mathbb{R}^{C_{\text{max}}}$, where C_{max} is the native maximum number of classes.
- For tasks with $C > C_{\text{max}}$, hierarchical classification trees compose node-wise probabilities by the chain rule.

2.5 Meta-Training with Synthetic Episodic Data

TabICL trains on synthetic tables drawn from a configurable prior, but each table is treated as a single supervised task. Few-shot structure is only partially encoded through attention masks, leading to (i) an **implicit** support–query signal and (ii) **uncontrolled** support selection, where updates may depend on redundant or uninformative examples.

Orion-Bix adopts an explicit *meta-learning* formulation on top of a similar synthetic prior. Synthetic tables are sampled over wide ranges of feature counts, label spaces, and distributions, and an `EpisodeGenerator` converts each table into many small support-query episodes. Episodes are created either by random splits or by kNN-based support selection that ensures both relevance and diversity. A `MetaLearningDataset` streams episodes in manageable chunks, and a `MetaLearningTrainer` processes thousands per update using micro-batching, gradient accumulation, mixed precision, and optional distributed data parallelism (DDP).

This episodic approach provides four advantages: (i) **alignment with the test-time objective**, since the model is always optimized in the regime in which it is evaluated; (ii) **richer task diversity**, as each synthetic table yields many distinct episodes; (iii) **higher-quality supports**, with kNN selection reducing redundant updates; and (iv) **scalable, stable optimization** enabled by micro-batching and distributed training.

2.6 Inference Pipeline and Practical Interface

Preprocessing and Feature Engineering. Given a table X , Orion-Bix preprocessor automatically: (i) detects numerical and categorical columns and converts all features to numeric form; (ii) imputes missing values for both types; (iii) applies an optional normalization scheme (none, power, quantile, or robust); (iv) clips outliers beyond a configurable z -score; and (v) generates alternative column orders via feature shuffling (e.g., circular shifts, random, Latin patterns).

Ensemble of Transformed Views. The preprocessor produces multiple transformed views of the dataset, each defined by a normalization choice and feature permutation. For each view, a support-query split is created and passed through Orion-Bix to obtain logits for the query rows. Across views, logits are class-aligned, averaged to form an ensemble prediction, and optionally converted to probabilities via temperature-scaled softmax. This multi-view ensembling improves stability and robustness, particularly under skewed distributions or strong feature-order sensitivity.

3 Experimental Evaluation

We evaluate Orion-Bix against TabICL and other baselines to highlight the benefits of biaxial attention and meta-learning. Our experiments focus on three aspects: (i) domain-specific performance on datasets with natural feature structure, (ii) support set quality to assess robustness, and (iii) few-shot learning curves across varying support sizes. These analyses target the improvements motivating Orion-Bix: structured multi-scale feature interactions and enhanced few-shot adaptation.

3.1 Evaluation Setup

3.1.1 Datasets. We construct domain-specific evaluation suites by grouping datasets from public benchmarks (e.g., **TALENT** [6] and **OpenML-CC18** [1]) according to application domain, allowing assessment in contexts where feature structure and heterogeneity are most relevant. The domains are: Medical Finance. These domains exhibit the high-dimensional, structured, and multi-scale features that Orion-Bix is designed to handle.

3.1.2 Evaluation Metrics. For each dataset, we report overall classification accuracy (ACC), class-weighted F1 to account for imbalance, and mean rank across datasets within each domain based on accuracy. All models use official train/test splits unless noted.

3.2 Domain-Specific Performance

Table 1: Domain-specific performance for Medical, Finance and Energy datasets from the benchmark suites. Formatting: Bold = 1st place; underlined = 2nd place within each group.

Models	Medical			Finance		
	Rank	ACC	F1	Rank	ACC	F1
XGBoost	6.32	0.7834	0.7669	6.62	0.7958	0.7885
RandomForest	6.38	0.7779	0.7752	7.32	0.8052	0.8001
CatBoost	6.36	0.7784	0.7594	<u>5.82</u>	0.8117	<u>0.8015</u>
LightGBM	5.32	0.7949	0.7614	6.17	0.8095	0.7974
TabICL	5.54	0.7819	0.7696	6.60	0.8125	0.7942
Orion-Bix	4.10	<u>0.7893</u>	<u>0.7759</u>	5.39	0.8206	0.8125
TabPFN	5.04	0.7984	0.7857	7.17	0.8094	0.7919
Mitra	10.77	0.3935	0.2863	13.67	0.5340	0.4250
ContextTab	8.66	0.6681	0.6129	11.25	0.7430	0.6834
TabDPT	6.86	0.7764	0.7641	8.00	0.8080	0.7960

Table 1 reports results on Medical, Finance, and Energy domains. Orion-Bix achieves the best mean rank in Medical (4.10) and Finance (5.39), outperforming TabICL and showing competitive performance against gradient-boosted baselines.

- **Medical.** Orion-Bix attains rank 4.10 (vs. TabICL 5.54), accuracy 0.7893, and F1 0.7759. While TabPFN slightly exceeds accuracy (0.7984), Orion-Bix’s superior ranking indicates more consistent performance across diverse datasets. The gains reflect the benefit of biaxial attention on grouped and hierarchical features typical in medical data.
- **Finance.** Orion-Bix leads with rank 5.39 (vs. TabICL 6.60), accuracy 0.8206 (vs. 0.8125), and F1 0.8125 (vs. 0.7942). Finance datasets feature strong hierarchies and multi-scale dependencies, where biaxial attention better captures structured relationships, notably improving F1 by +1.83 points over TabICL.

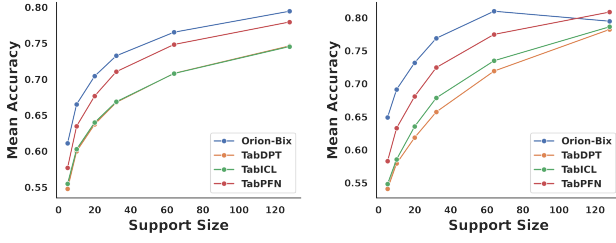
These results highlight two design benefits of Orion-Bix:

- (1) **Biaxial attention for structured features:** Gains are largest in domains with natural feature groups and hierarchies, confirming that modeling local (grouped), coarse-scale (hierarchical), and global (relational) interactions improves representations over a single attention stack.
- (2) **Multi-CLS aggregation:** Maintaining multiple CLS tokens preserves distinct feature aspects, capturing both rare patterns and common global signals, leading to more robust predictions.

3.3 Few-Shot Performance

We assess performance as a function of support set size, $k \in \{5, 10, 20, 32, 64, 128\}$, to test whether Orion-Bix’s meta-learning benefits are most pronounced in the very few-shot regime.

Protocol. For each dataset and support size k , k examples are sampled from the training set (all classes represented) to form the support set. Models are evaluated on the full test set, repeated across



(a) Accuracy vs Support Size on TALENT Benchmark. (b) Accuracy vs Support Size on OpenML-CC18 Benchmark.

Figure 1: Accuracy of Orion-Bix and TabICL across different support sizes in few-shot experiments.

multiple seeds, and results are averaged. Accuracy is then plotted against support size.

Results. Figure 1 shows:

- Orion-Bix attains the best accuracy for very low shot sizes ($k \leq 32$), outperforming all three baselines by 2–4 points, indicating meta-training on diverse episodes improves data efficiency.
- Accuracy increases monotonically with k for all models; gains are largest from $k = 5$ to $k = 32$, while improvements beyond $k = 64$ are small, indicating diminishing returns from additional support examples.
- As k grows, the gap narrows: on TALENT, Orion-Bix stays clearly best across all k , while on OpenML-CC18 TabPFN continues improving and slightly surpasses Orion-Bix at $k = 128$. In both cases, TabPFN and Orion-Bix remain well ahead of TabICL and TabDPT. Overall, Orion-Bix is strongest in the few-shot regime.

3.4 Support Set Quality Analysis

We study how support set selection affects few-shot performance by comparing random vs. diverse (kNN-based) sampling *at inference*. This tests whether Orion-Bix’s meta-learning, which constructs diverse support/query episodes during pre-training, improves robustness to support quality.

Protocol. Each test dataset is split 80/20 into train/test sets. A support set of size $k = 32$ is drawn using: (i) **Uniform sampling**, ensuring all classes are represented, and (ii) **Diverse (kNN-based)**, selecting examples close to test queries and maximally diverse. Models are evaluated with standard ICL on the same support/query splits, repeated across multiple seeds for statistical significance.

Motivation. This protocol measures *inference-time robustness*: diverse selection simulates a scenario where practitioners can actively choose informative support examples, while random selection represents default, arbitrary support sampling.

Results. Figure 2 shows that: (1) Uniform sampling slightly outperforms kNN-based selection (by 3–5 points), indicating few-shot performance is not highly sensitive to support curation; (2) Orion-Bix consistently exceeds TabICL under both schemes, with relative

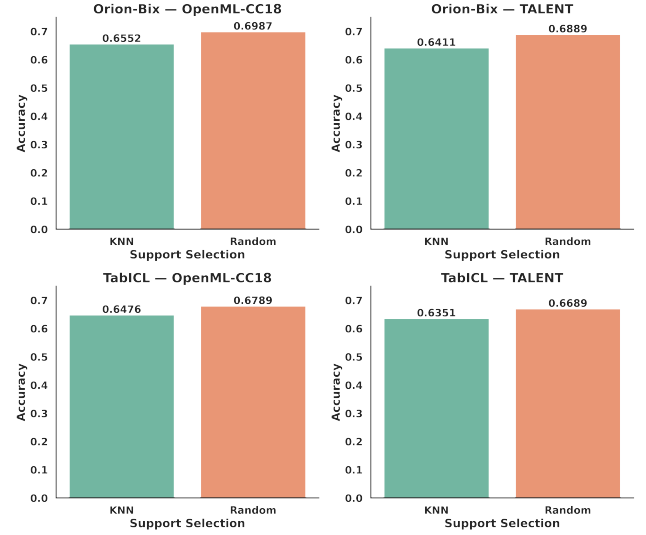


Figure 2: Accuracy of Orion-Bix and TabICL on OpenML and TALENT benchmarks. Bars indicate support selection strategies, averaged over datasets for each model.

ordering unchanged, suggesting meta-training improves overall in-context generalization rather than depending on a specific support selection heuristic.

4 Conclusion

In summary, Orion-Bix preserves the core strengths of TabICL while adding a biaxial row encoder and an episodic meta-learning schedule that directly targets few-shot in-context performance.

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